



Lab Task 05

Submitted By	Ijlal Ahmad
Submitted To	Dr. Jaweria Kanwal
Date	22-03-2025
Subject	SCD

1. Code

```
ZakatCalculator.java ×
1 package project1.module1;
2
3 public class ZakatCalculator {
4
5
6     private static final double ZAKAT_PERCENTAGE = 0.025;
7
8
9     public double calculate_zakat(double total_savings) {
10         if (total_savings < 0) {
11             throw new IllegalArgumentException("Total savings cannot be negative.");
12         }
13         return total_savings * ZAKAT_PERCENTAGE;
14     }
15
16     public static void main(String[] args) {
17         ZakatCalculator calculator = new ZakatCalculator();
18
19
20         double total_savings = 10000.0;
21         double zakat_amount = calculator.calculate_zakat(total_savings);
22
23         System.out.println("Total Savings: $" + total_savings);
24         System.out.println("Zakat Payable: $" + zakat_amount);
25     }
26 }
```

Output

```
Total Savings: $10000.0
Zakat Payable: $250.0
```

2. Code

```
1 package project1.module1;
2
3 public class CarRentalSystem {
4
5     private String carModel;
6     private String carLicensePlate;
7     private int carYear;
8
9     private int rentalDurationDays;
10
11     private String customerName;
12     private String customerPhoneNumber;
13
14     public CarRentalSystem(String carModel, String carLicensePlate, int carYear, int rentalDurationDays,
15                             String customerName, String customerPhoneNumber) {
16         this.carModel = carModel;
17         this.carLicensePlate = carLicensePlate;
18         this.carYear = carYear;
19         this.rentalDurationDays = rentalDurationDays;
20         this.customerName = customerName;
21         this.customerPhoneNumber = customerPhoneNumber;
22     }
23
24     public String getCarModel() {
25         return carModel;
26     }
```

```
    public void setCarModel(String carModel) {
        this.carModel = carModel;
    }
```

```
    public String getCarLicensePlate() {
        return carLicensePlate;
    }
```

```
    public void setCarLicensePlate(String carLicensePlate) {
        this.carLicensePlate = carLicensePlate;
    }
```

```
    public int getCarYear() {
        return carYear;
    }
```

```
    public void setCarYear(int carYear) {
        this.carYear = carYear;
    }
```

```
    public int getRentalDurationDays() {
        return rentalDurationDays;
    }
```

```

52
53 public void setRentalDurationDays(int rentalDurationDays) {
54     this.rentalDurationDays = rentalDurationDays;
55 }
56
57
58 public String getCustomerName() {
59     return customerName;
60 }
61
62 public void setCustomerName(String customerName) {
63     this.customerName = customerName;
64 }
65
66 public String getCustomerPhoneNumber() {
67     return customerPhoneNumber;
68 }
69
70 public void setCustomerPhoneNumber(String customerPhoneNumber) {
71     this.customerPhoneNumber = customerPhoneNumber;
72 }
73
74
75 public void printDetails() {
76     System.out.println("Car Details:");
77     System.out.println("Model: " + carModel);
78
79     public void printDetails() {
80         System.out.println("Car Details:");
81         System.out.println("Model: " + carModel);
82         System.out.println("License Plate: " + carLicensePlate);
83         System.out.println("Year: " + carYear);
84         System.out.println("\nRental Duration:");
85         System.out.println("Days: " + rentalDurationDays);
86         System.out.println("\nCustomer Information:");
87         System.out.println("Name: " + customerName);
88         System.out.println("Phone Number: " + customerPhoneNumber);
89     }
90
91     public static void main(String[] args) {
92         CarRentalSystem rental = new CarRentalSystem(
93             "Toyota Camry",
94             "ABC-1234",
95             2020,
96             7,
97             "John Doe",
98             "123-456-7890"
99         );
100
101         System.out.println("Initial Rental Details:");
102         rental.printDetails();

```

```
29         rental.printDetails();
30
31         rental.setRentalDurationDays(10);
32         rental.setCustomerName("Jane Smith");
33
34         System.out.println("\nUpdated Rental Details:");
35         rental.printDetails();
36     }
37 }
```

Output

```
Initial Rental Details:
Car Details:
Model: Toyota Camry
License Plate: ABC-1234
Year: 2020

Rental Duration:
Days: 7

Customer Information:
Name: John Doe
Phone Number: 123-456-7890

Updated Rental Details:
Car Details:
Model: Toyota Camry
License Plate: ABC-1234
Year: 2020

Rental Duration:
Days: 10

Customer Information:
Name: Jane Smith
Phone Number: 123-456-7890
```